Downscaling Municipal Soybean Yield Labels to Pixel Predictions with MSTNet in Paraná, Brazil

Abstract

Official crop yield statistics are often reported at the level of administrative units, such as municipalities in Brazil and counties in the United States, rather than at finer spatial resolutions. However, precision agriculture, insurance, and localized risk assessment require estimates at finer spatial resolutions. We propose MSTNet (*Multi* STNet), a Transformer model that combines data sampled at different temporal frequencies. Separate STNet trunks encode soy-masked Sentinel-2 time series and a daily gridded climate series, while annual soil properties enter the decoder by late fusion. Since thehe model is able to predict one value per soy pixel, we can use it to predict yield for arbitrary polygons. Training uses municipality-mean Sentinel-2 series under PAM supervision; at evaluation the same weights score every soy pixel and average to municipalities, so arbitrary polygons can be formed at inference without retraining. We evaluate in Paraná, Brazil, with year leave-one-out over harvest years 2020–2024, reporting only the left-out year in each fold. Excluding the 2022 drought fold, mean LOO R^2 is 0.302; when 2022 itself is left out, performance collapses because no comparable drought year remains in training. Direct comparison with earlier work is limited for this municipal-supervision, pixel-scoring setting; we report the LOO benchmark and leave field-referenced checks for future work.

1 Introduction

Reliable soybean productivity estimates matter for agricultural monitoring, insurance, and risk assessment. In Brazil, the Municipal Agricultural Production survey (PAM) from the Brazilian Institute of Geography and Statistics (IBGE) publishes official yields only at the municipality level. The same kind of coarse reporting is common elsewhere. Sentinel-2 imagery, gridded weather, and soil maps describe crop condition at much finer scales, but field-level yield labels are rarely available statewide. What we need is a model that can learn from municipal labels and still make predictions for smaller regions, at a scale that is cheap enough to run over large areas.

MSTNet (*Multi* STNet) is our answer to that requirement. It extends the Spectral–Temporal Network (STNet) of Xu et al. [2024] by giving each covariate stream its own trunk when the streams are sampled at different frequencies. Here, one trunk encodes soy-masked Sentinel-2 reflectance and another encodes daily climate. Annual MapBiomas Solo properties, which are static once the harvest year is fixed, are concatenated in the decoder. Predictions are defined per soy pixel. For speed, training fits the network on municipality-mean Sentinel-2 series supervised by PAM. At evaluation and inference we apply the same weights to every soy pixel and average the scores to municipalities (or to any other polygon) without retraining.

We run the experiments on soybean municipalities in Paraná for harvest years 2020–2024 under year leave-one-out: each fold trains on four years and scores only the left-out year. Because PAM is the only supervised target, the numbers we report are municipal aggregates of pixel scores on those LOO years. We are not aware of many yield studies that train a multi-frequency Transformer on municipal labels while still exposing pixel scores for downscaling. Matched baselines for that setup are scarce, so we present the LOO benchmark rather than a direct comparison against the literature.

Table 1: Summary of data sources used in the experiments.

Category	Variables and processing	Source
Satellite	Ten Sentinel-2 L2A reflectance bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12), soy-masked zonal means via Google Earth Engine	Copernicus Sentinel-2 L2A [Copernicus Sentinel-2, processed by ESA, 2021]; GEE [Gorelick et al., 2017]
Climate	Daily precipitation, ETo, Rs, Tmax, and Tmin; averaged across Xavier cells within the municipality polygon	Xavier BR-DWGD [Xavier et al., 2022]
Soil	Clay, silt, sand, and soil organic-carbon stock at 0–30 cm; late-fused static covariates	MapBiomas Solo Collection 3 [MapBiomas, 2025b]
Crop mask	MapBiomas soybean class 39, used to select pixels before Sentinel-2 aggregation	MapBiomas Collection 10 [MapBiomas, 2025a]
Target	Municipal soybean productivity (t/ha)	IBGE PAM [Instituto Brasileiro de Geografia e Estatística, 2026]

Our contributions are MSTNet itself, the pixel-scoring setup that turns municipal supervision into sub-municipal maps, and a five-fold year-LOO evaluation for Paraná over 2020–2024.

2 Data

The analysis uses municipal PAM productivity labels together with Sentinel-2, daily climate, and soil covariates (Table 1). Soy-masked Sentinel-2 pixels provide the spectral signal. Climate is reduced to a municipal daily series. Soil can be attached per pixel or as a municipal vector. Figure 1 shows the municipality-mean spectral path used for training. Appendix A shows the pixel-by-pixel path used for the reported LOO metrics and for sub-municipal maps.

2.1 Study area and labels

The study covers soybean-producing municipalities in the state of Paraná, Brazil. The target is municipal soybean productivity in tonnes per hectare (t/ha) from IBGE PAM [Instituto Brasileiro de Geografia e Estatística, 2026]; municipality code and harvest year identify each observation. No field- or pixel-level yield labels are used. We retain municipality–year pairs whose Sentinel-2 coverage ratio relative to the reported planted area lies in $[0.5, 1.5]$, so that satellite coverage is roughly consistent with official area statistics.

2.2 Covariates and municipal aggregation

Sentinel-2 L2A reflectance is processed in Google Earth Engine [Gorelick et al., 2017]: for each municipality and acquisition date we compute soy-masked zonal means of ten bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12), using MapBiomas Collection 10 class 39 as the soybean mask [MapBiomas, 2025a]. Each municipal spectral sequence carries a day-of-year index for positional encoding and is truncated or padded to $T = 45$ acquisitions. Climate covariates come from the Xavier daily weather grid [Xavier et al., 2022] as a separate municipal daily series over the inclusive Oct 1–Mar 31 window (length 183): precipitation, reference evapotranspiration (ETo), solar radiation (Rs), maximum temperature (Tmax), and minimum temperature (Tmin). Daily climate is averaged over Xavier cells whose centers fall in the municipality polygon and is scaled independently of reflectance; MSTNet encodes it in a dedicated trunk rather than concatenating climate onto Sentinel-2 timesteps. Static MapBiomas Solo properties (clay, silt,

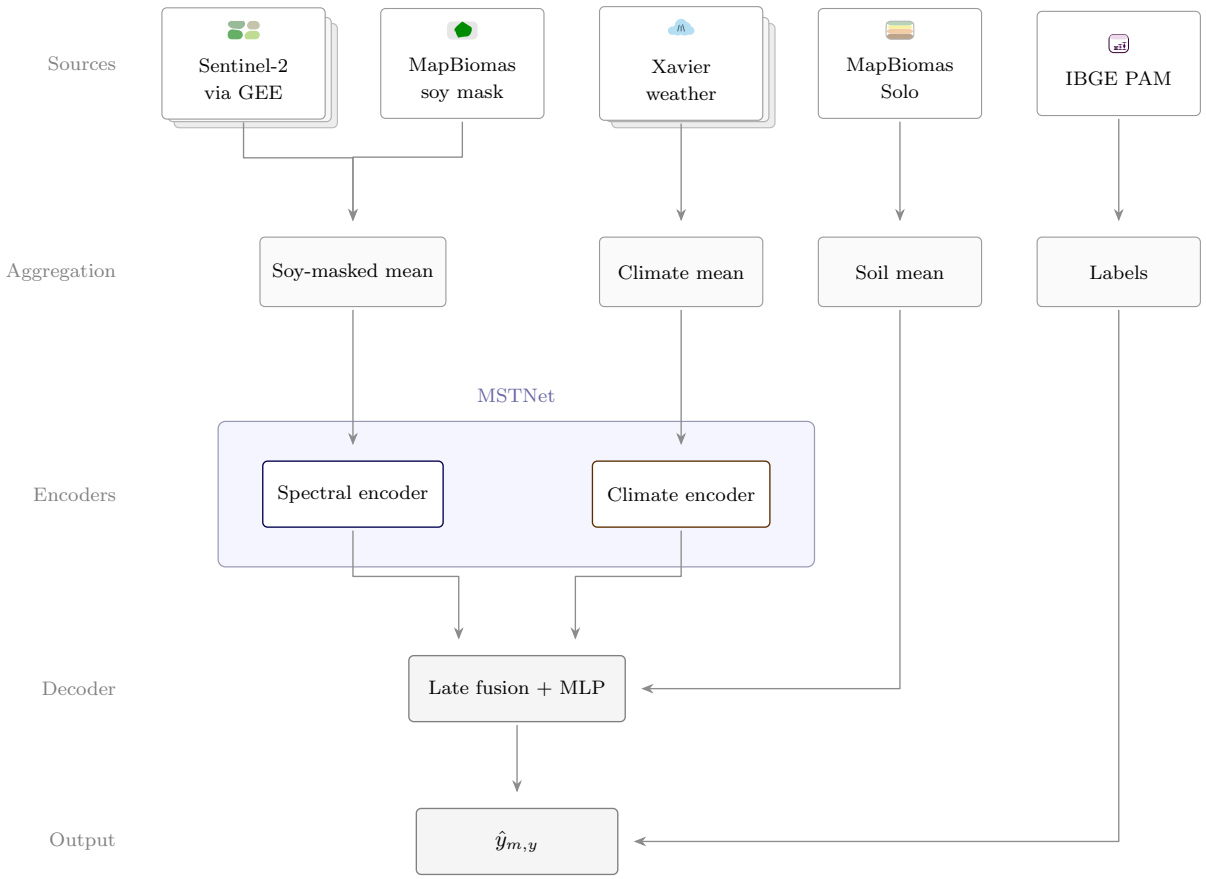


Figure 1: Data pipeline for MSTNet training. Sentinel-2 is reduced to soy-masked municipal means; Xavier weather is encoded by a parallel trunk; MapBiomas Solo enters the decoder by late fusion. The decoder prediction $\hat{y}_{m,y}$ is supervised by IBGE PAM labels at the output. Reported evaluation instead scores individual soy pixels and averages them (Appendix A).

sand, and SOC stock at 0–30 cm) are late-fused in the decoder [MapBiomas, 2025b]. An ablation omits climate and soil inputs and uses the ten spectral bands alone.

2.3 Year leave-one-out splits

Evaluation follows year leave-one-out over harvest years 2020–2024 (Section 4.2). In each fold, every municipality–year from the four training years is used for fitting; the left-out year is never seen in training and provides the only reported municipal scores for that fold. Target standardization is recomputed from each fold's training years. All main-text metrics are municipal aggregates obtained by averaging pixel-level scores on the LOO year only (Section 3.1). We do not validate pixel maps against field yields.

3 Methods

3.1 Pixel scoring under municipal supervision

Let $\mathcal{P}_{m,y}$ denote the soybean pixels observed in municipality m in harvest year y. MSTNet defines a productivity score $\hat{y}_p$ for each soy pixel p from its Sentinel-2 sequence $\mathbf{x}_p$, the municipal daily climate series $\mathbf{c}_{m,y}$ (broadcast to all pixels in m), and soil covariates $\mathbf{s}_p$ (or a municipal soil vector when only polygon-level soil is available). Reported municipal predictions average those

pixel scores,

$$\hat{y}_{m,y} = \frac{1}{|\mathcal{P}_{m,y}|} \sum_{p \in \mathcal{P}_{m,y}} \hat{y}_p, \qquad \hat{y}_p = f_\theta(\mathbf{x}_p, \mathbf{c}_{m,y}, \mathbf{s}_p), \tag{1}$$

and are compared with the PAM label $y_{m,y}$. At inference the field $\{\hat{y}_p\}$ can also be kept as a map or averaged over any sub-municipal polygon. Appendix A sketches this pipeline.

Training uses a cheaper surrogate: soy-masked Sentinel-2 reflectance is averaged before encoding,

$$\bar{\mathbf{x}}_{m,y,t} = \frac{1}{|\mathcal{P}_{m,y,t}|} \sum_{p \in \mathcal{P}_{m,y,t}} \mathbf{x}_{p,t}, \tag{2}$$

so each municipality–year is one forward pass $\hat{y}^{\text{train}}_{m,y} = f_\theta(\bar{\mathbf{x}}_{m,y}, \mathbf{c}_{m,y}, \mathbf{s}_m)$ supervised by PAM. The LOO metrics in Sections 4–5 are not from that surrogate: they apply the trained weights pixel by pixel and aggregate with (1). Because f_θ is nonlinear, mean-then-encode and encode-then-mean need not coincide; the pixel-mean evaluation is the operational downscaling check. Appendix B reports the same LOO years under the municipal-mean evaluation path for reference.

3.2 Architecture

MSTNet (*Multi* STNet) places one STNet trunk on each time-varying stream so that modalities with different sampling rates need not share a common calendar [Xu et al., 2024]. A spectral trunk encodes Sentinel-2 sequences (ten bands, length $T = 45$) with day-of-year positional encodings and NDVI-weighted temporal pooling. A climate trunk encodes daily Xavier weather (five channels: precipitation, ETo, Rs, Tmax, Tmin) over the Oct–Mar window (length 183), pooled by a mean over valid days. MapBiomas Solo properties are fixed once the harvest year is known. They are concatenated with the pooled trunk embeddings and passed through a multilayer-perceptron decoder that outputs a productivity estimate (per pixel or per municipal aggregate, depending on the input path). As an ablation on a spectral-only single-trunk STNet, we also try mean pooling over valid spectral timesteps instead of NDVI weights. That spectral-only model uses ten-band inputs and has no climate or soil branches.

Each MSTNet trunk uses one Transformer layer, $d_{\text{model}} = 128$, four attention heads, feed-forward width $d_{\text{inner}} = 64$, and dropout 0.4.

3.3 Loss and optimization

Training minimizes mean squared error between predicted and observed municipal productivity after standardizing targets with the training-set mean and standard deviation. Reported metrics (RMSE, MAE, and R^2) are computed after transforming predictions back to t/ha.

Optimization uses AdamW with learning rate 1×10^{-3}, weight decay 5×10^{-4}, and batches of 40 municipalities. The learning rate is scheduled to drop at epochs 40 and 120; training stops early after fifteen epochs without improvement in validation R^2. No self-supervised pretraining is used: the network is trained from scratch under municipal supervision.

4 Experiments

4.1 Proposed model

The proposed model is MSTNet: one STNet trunk for Sentinel-2, one for daily Xavier climate (precipitation, ETo, Rs, Tmax, and Tmin), and late fusion of MapBiomas Solo properties in the decoder. Each trunk uses one Transformer layer with $d_{\text{model}} = 128$, four attention heads, $d_{\text{inner}} = 64$, and dropout 0.4. Training fits municipality-mean Sentinel-2 series; reported metrics apply the selected weights pixel by pixel and average the scores to municipalities (Section 3.1).

Table 2: Year leave-one-out metrics for MSTNet (pixel-by-pixel scores averaged to municipalities). Each row is a fold that trains on the other four harvest years and evaluates only the listed LOO year. RMSE and MAE are in t/ha.

LOO year	n	RMSE	MAE	R^2
2020	346	0.395	0.316	0.291
2021	343	0.323	0.244	0.402
2022	342	1.404	1.136	−0.422
2023	339	0.424	0.337	0.231
2024	336	0.568	0.439	0.284
Mean	—	0.623	0.495	0.157
Mean excl. 2022	—	0.427	0.334	0.302

4.2 Year leave-one-out design

We evaluate temporal generalization with year leave-one-out (LOO) over harvest years 2020–2024. For each fold with left-out year Y:

- all municipality–years with harvest year $\neq Y$ are used for training;
- year Y is held out entirely (no municipality from Y enters training);
- model selection within the fold uses validation R^2 on year Y;
- the reported fold score is the pixel-mean municipal evaluation on year Y only.

Input standardization is refit on each fold's training years. This yields five independently trained checkpoints, one per left-out year. Main-text tables and figures report only those LOO-year scores. Appendix B records the municipal-mean evaluation path on the same LOO years.

4.3 Evaluation protocol

For each LOO fold we compare municipal aggregates of pixel predictions with PAM labels on the held-out year. We report per-fold RMSE, MAE, and R^2, predicted-versus-observed scatterplots for every LOO year, a municipal error map for a non-drought fold (2021), and a qualitative intramunicipal map for Guarapuava.

5 Results

5.1 Year-LOO municipal accuracy

Table 2 reports pixel-mean municipal metrics on the left-out year of each fold. Across the five folds, mean R^2 is 0.157 and mean RMSE is 0.623 t/ha. Excluding the 2022 drought fold, mean R^2 rises to 0.302 and mean RMSE falls to 0.427 t/ha. The strongest non-drought fold is 2021 ($R^2 = 0.402$, RMSE = 0.323 t/ha).

5.2 Drought year 2022

Harvest 2022 is the only year in our 2020–2024 sample with a severe drought across Paraná. When 2022 is the LOO year, the training set contains no analogous drought season, so the model must extrapolate outside the climatic regimes seen in training. That fold collapses: $R^2 = -0.422$ and RMSE = 1.404 t/ha, far worse than the other LOO years. Figure 2 shows the corresponding municipal error map. The remaining folds, trained with 2022 available when another year is

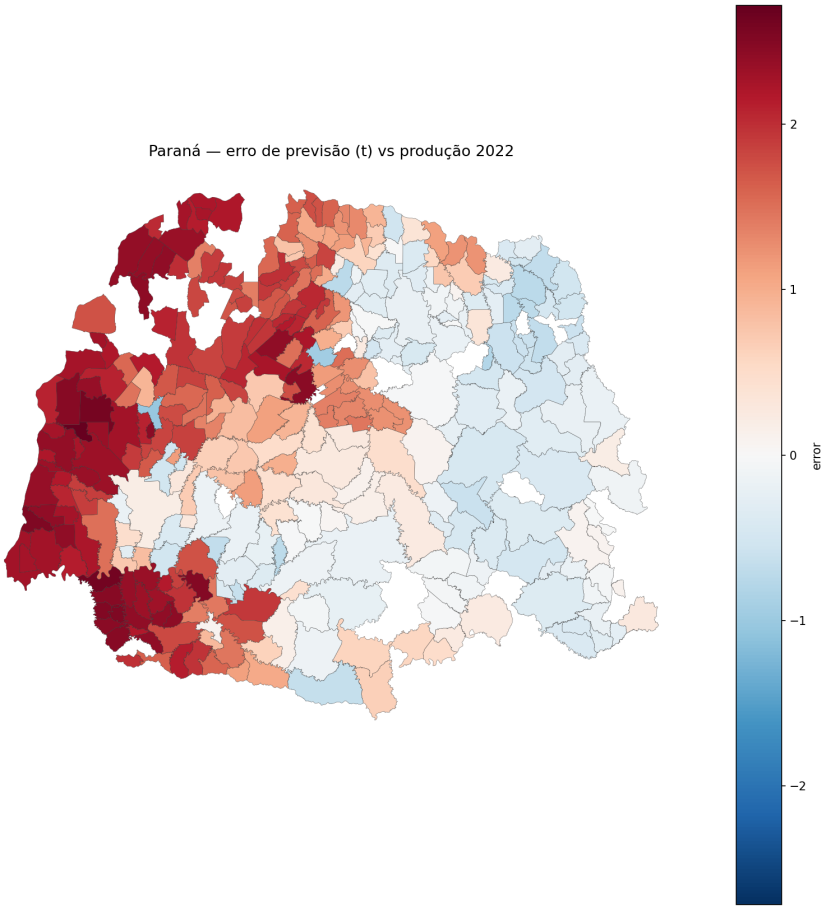


Figure 2: Municipal aggregate prediction error map for LOO year 2022 (drought fold).

held out, do not face that missing-regime problem to the same degree. We therefore treat 2022 as a stress test of out-of-distribution seasons rather than as a typical LOO score, and we report means both with and without that fold.

5.3 Scatterplots and spatial examples

Figure 3 shows predicted versus observed municipal productivity for every LOO year. Non-drought folds track the identity line with moderate scatter; the 2022 panel shows the systematic failure under drought. Figure 4 maps municipal errors for the 2021 LOO fold. Figure 5 shows pixel-level scores for Guarapuava under the 2021 LOO checkpoint. That map is qualitative; we have no field yields to score it against, only the municipal PAM total when the pixels are averaged.

6 Discussion

Year leave-one-out gives a clearer picture of temporal risk than a single fixed holdout. On non-drought folds, municipal aggregates of pixel scores reach R^2 between 0.23 and 0.40, with RMSE about 0.32–0.57 t/ha. The pooled mean excluding 2022 is $R^2 = 0.302$ and RMSE $= 0.427$ t/ha.

The 2022 fold is the exception that matters for operations. It is the only drought year in the sample, so leaving it out removes every training example from that regime. The resulting negative R^2 and RMSE above 1.4 t/ha show that the present covariates and capacity do not invent drought behaviour from wetter years alone. When 2022 remains in training and another

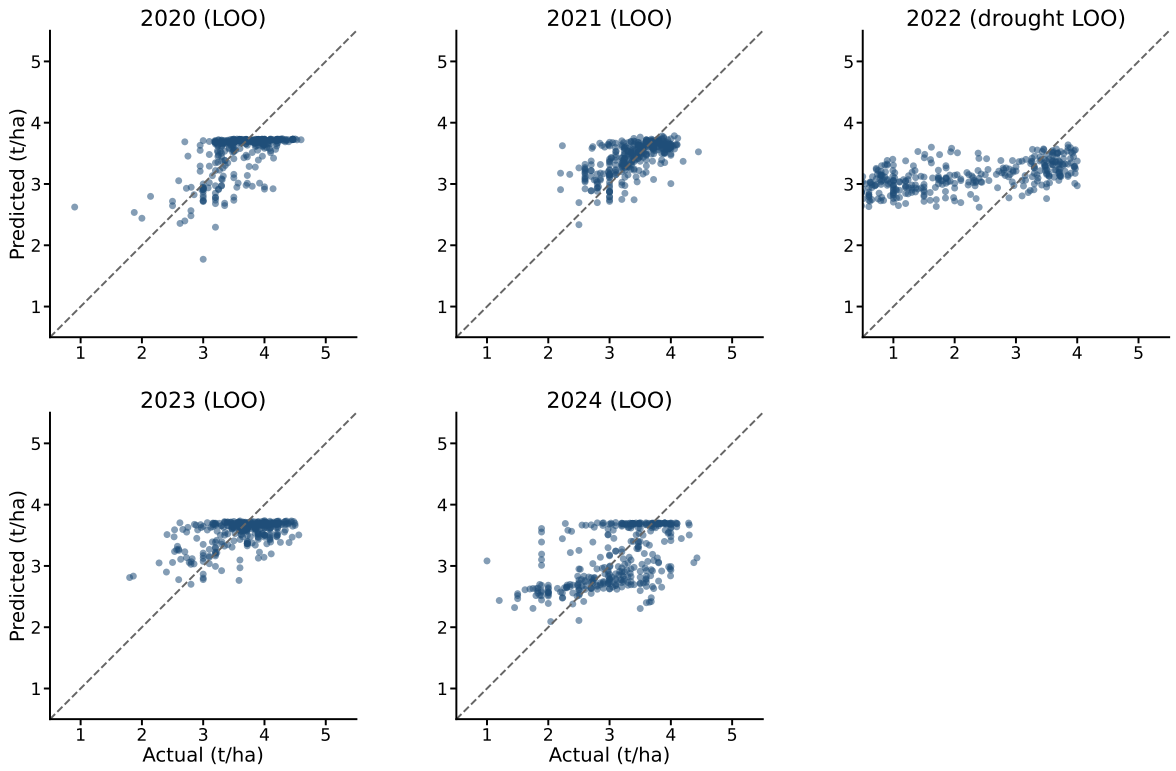


Figure 3: Predicted versus observed municipal soybean productivity (t/ha) for each year-LOO fold of MSTNet. Every panel evaluates only the left-out year. The dashed line is identity.

year is held out, LOO skill is much more stable. Any operational deployment should therefore monitor season type explicitly and avoid treating a single mild year as representative.

Because scores are defined per pixel, municipal PAM labels can still support downscaling to arbitrary sub-municipal shapes (Figure 5). The numbers here evaluate municipal aggregates of those pixel scores; they do not check intramunicipal maps against field yields. Training used municipality-mean Sentinel-2 series for speed, so LOO skill is a transfer check under encode-then-mean evaluation. Other limitations are the single-state geography, no repeated seeds, and scarce matched external baselines. Next steps are multi-state LOO, controlled modality ablations, stronger baselines, and independent yield data when spatially resolved accuracy is required.

7 Conclusion

We presented MSTNet, a multi-frequency Transformer that uses Sentinel-2 time series, daily Xavier climate, and MapBiomas Solo covariates to predict soybean productivity from municipal PAM labels while scoring soy pixels for sub-municipal maps. Under year leave-one-out on Paraná harvest years 2020–2024, municipal aggregates of pixel scores reach mean $R^2 = 0.302$ and RMSE $= 0.427\,\mathrm{t/ha}$ when the 2022 drought fold is excluded; that drought LOO fold fails ($R^2 = -0.422$) because no similar season remains in training.

These figures give a reproducible LOO benchmark for municipal-supervision, pixel-scoring yield models in Paraná. Calibrated intramunicipal maps, broader geographies, and claims about individual covariate groups still need field checks, matched ablations, and additional LOO seasons.

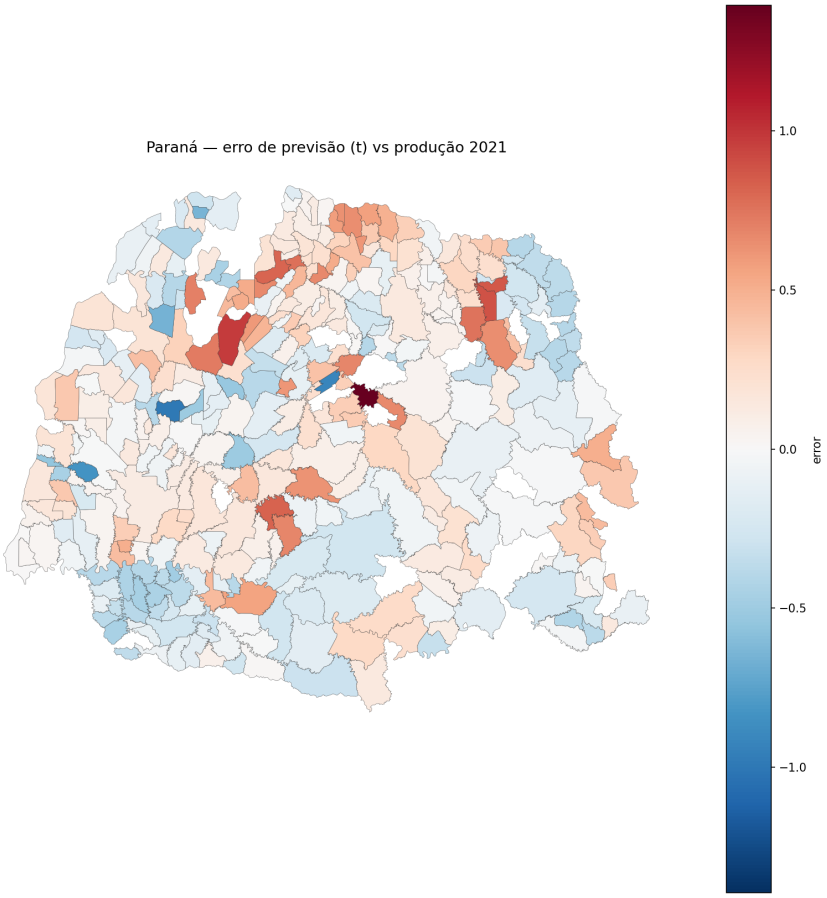


Figure 4: Municipal aggregate prediction error map for LOO year 2021.

Data availability

Municipal productivity labels are from IBGE PAM and provide the only supervised signal. Sentinel-2 L2A reflectance was processed in Google Earth Engine with MapBiomas soybean masking; daily Xavier climate and MapBiomas Solo enter as described in Section 2. Training uses municipality-mean Sentinel-2 series; reported evaluation scores soy pixels and averages to municipalities (Sections 3 and A). Further details are available from the authors upon reasonable request.

A Pixel-level MSTNet pipeline

Section 3.1 defines the pixel-scoring evaluation (1): soy-masked Sentinel-2 pixels are scored one by one, municipal climate is shared across pixels, soil enters by late fusion, and the mean of the pixel predictions is compared with PAM. Figure 1 shows the municipality-mean spectral path used for training. Figure 6 shows the pixel-by-pixel path used for the reported LOO metrics and for intramunicipal maps. At inference the pixel field $\{\hat{y}_p\}$ can be kept as a map or averaged over any polygon. That map is not a training target.

B Municipal-mean evaluation on LOO years

Training encodes soy-masked municipal-mean Sentinel-2 series (2). Table 2 in the main text instead scores every soy pixel and averages the predictions (1). Table 3 reports the same left-out years under the municipal-mean evaluation path, using the same fold checkpoints. The two paths

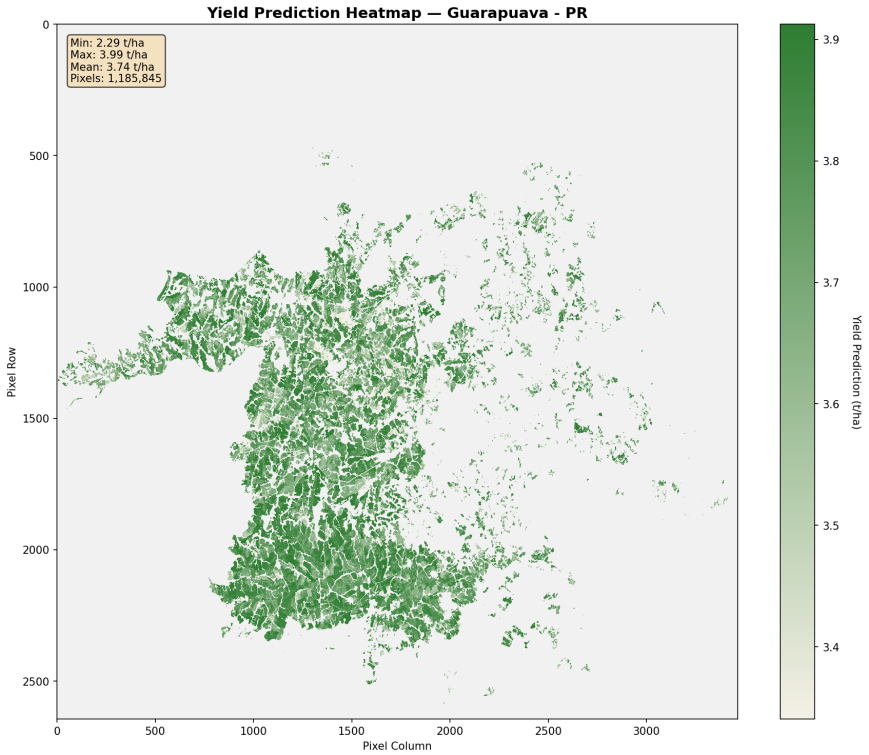


Figure 5: Intramunicipal map of downscaled pixel-level productivity predictions for Guarapuava, Paraná (LOO year 2021).

are close on non-drought folds; the 2022 drought fold remains poor under either aggregation ($R^2 = -0.300$ municipal-mean versus -0.422 pixel-mean). Excluding 2022, mean municipal-mean LOO R^2 is 0.275.

Table 3: Year leave-one-out metrics for MSTNet under the municipal-mean evaluation path (mean-then-encode). Each row evaluates only the listed LOO year with the fold trained on the other four harvest years. RMSE and MAE are in t/ha.

LOO year	n	RMSE	MAE	R^2
2020	346	0.411	0.316	0.233
2021	343	0.320	0.239	0.410
2022	342	1.342	1.069	−0.300
2023	339	0.433	0.345	0.199
2024	336	0.577	0.445	0.259
Mean	—	0.617	0.483	0.160
Mean excl. 2022	—	0.435	0.336	0.275

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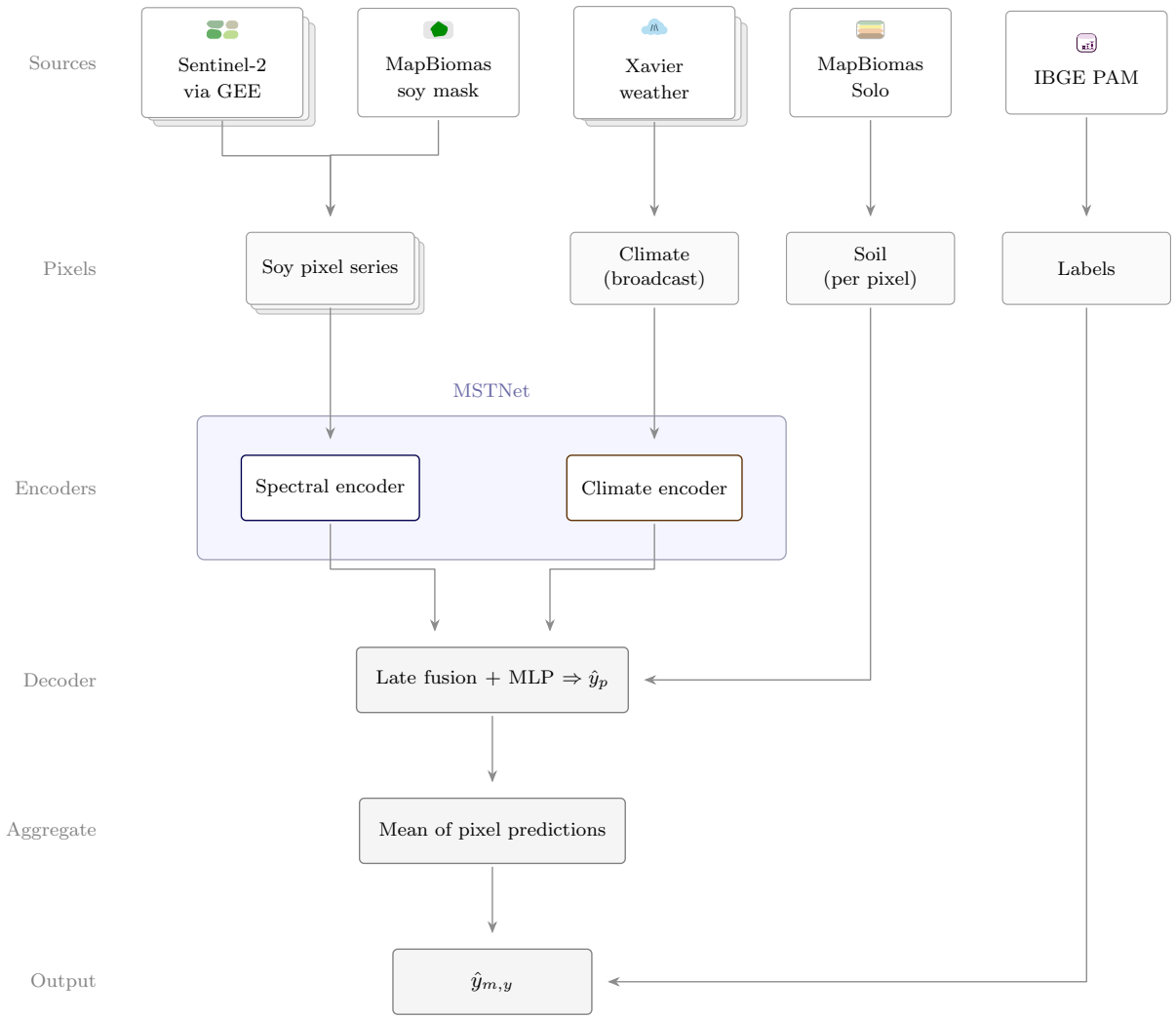


Figure 6: Pixel-by-pixel MSTNet pipeline. Soy-masked Sentinel-2 pixels are encoded individually; municipal Xavier climate is broadcast to pixels and soil enters by late fusion. The mean of pixel predictions $\hat{y}_p$ yields $\hat{y}_{m,y}$, which is supervised by IBGE PAM.

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